

Real-Time AI/NLP Financial Risk Engine & Wholesale Portfolio Stress Tester

CRISIL / S&P Challenge Submission

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Traditional risk signals arrive after the market has moved

LIVE INTELLIGENCE LOOP



Business Problem

Traditional institutional risk management relies on lagging numerical reporting cycles, missing early-warning signals of liquidity crises, central bank rate shifts, or geopolitical shocks buried in unstructured news and social feeds.

Solution Approach

A unified end-to-end intelligence pipeline bridging natural language processing (NLP) with quantitative Basel III wholesale portfolio stress testing to automate market impact simulations.

One intelligence loop connects live signals to portfolio stress

01

Data Ingestion Layer

Multi-source live connectors aggregating GDELT 2.0, NewsAPI, Finnhub, and financial RSS wires.

02

AI/NLP Intelligence Unit

ProsusAI/FinBERT transformer model paired with financial lexicons and automated coordinate geocoding.

03

Downstream Module B

Event-driven stress testing engine monitoring a $278.5M USD Exposure at Default (EAD) wholesale portfolio.

04

Presentation Tier

Unified Streamlit analytical dashboard and a React/Vite 3D WebGL frontend.

Resilience is designed into the live-demo path



Resilient NLP Pipeline: Lazy-loaded transformer architecture with automated fallback tiers to guarantee zero-crash execution during live demonstrations.



Geospatial Intelligence: Translates unstructured text entities into precise latitude/longitude coordinates to map active risk origins.



Institutional Tech Stack: Asynchronous FastAPI backend, Pydantic V2 schema validation, and modular data exchange via JSON payloads.

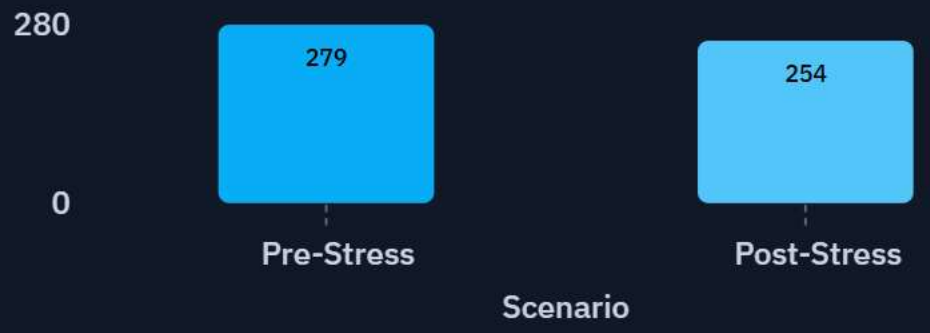
High-severity credit events compress value and amplify expected loss

Structured Output

Converts raw text into standardized JSON records containing sentiment polarity, event taxonomy, and impact scores.

Asset-Level Valuation Shock (Pre vs. Post Stress)

Wholesale portfolio drops from $278.50M to $253.52M (-8.97% MTM) under high-severity credit events.



Expected Loss (EL) Credit Risk Surge by Asset Class

Spikes across High-Yield Bonds, Commercial Real Estate, and Corporate Loans.

Asset Class	Expected Loss Surge
High-Yield Bonds	High
Commercial Real Estate	High
Corporate Loans	High

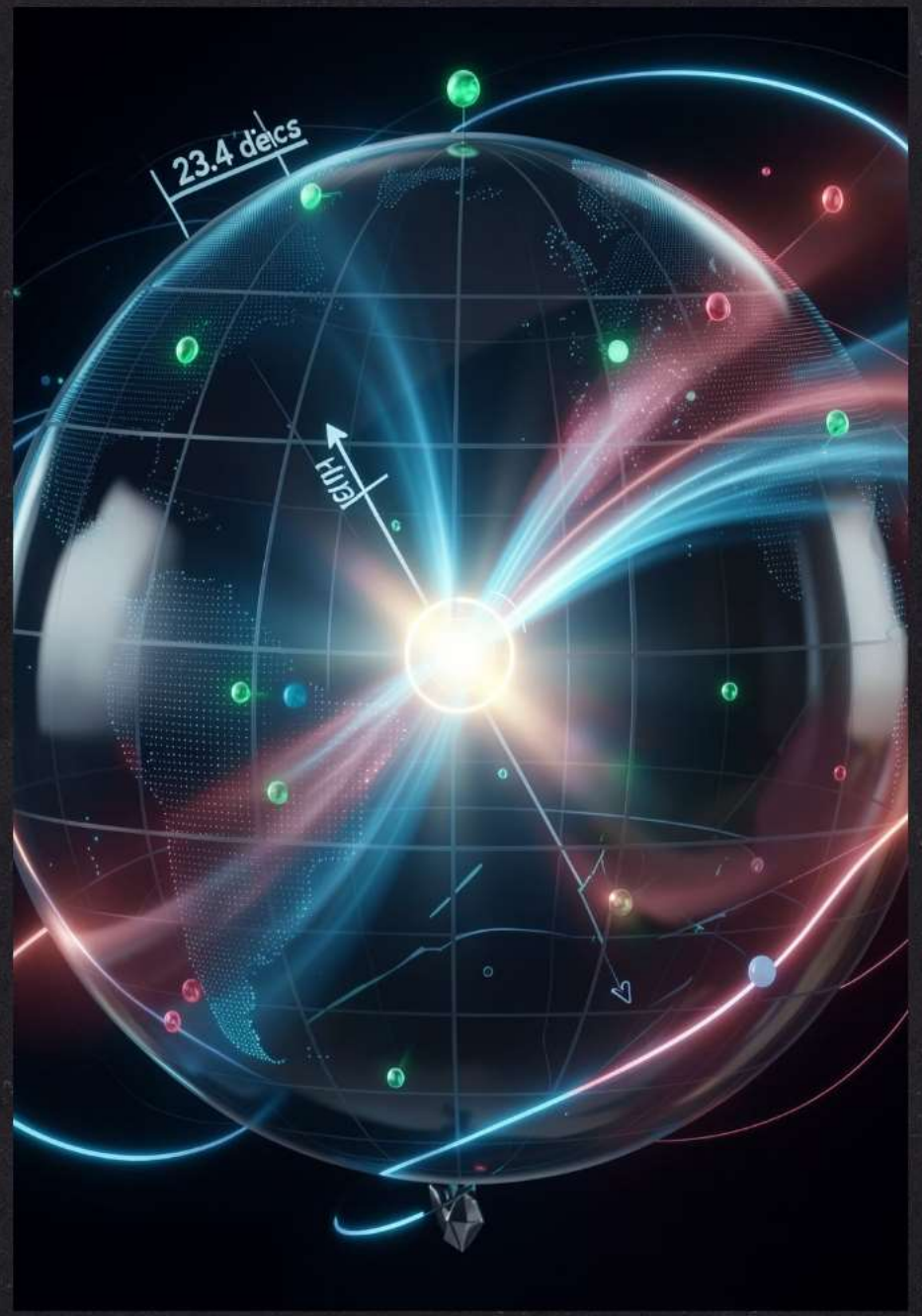
3D risk geosphere turns global events into a portfolio view

Strategic Value

Shifts risk management from reactive reporting to real-time predictive monitoring, automating Basel III compliance workflows.

Innovation Showcase (3D WebGL Risk Globe)

Custom Three.js interactive risk geosphere featuring dotted continental landmasses, latitude/longitude grid lines, 23.4° axial tilt, real-time mouse drag controls, and animated contagion flow arcs targeting the Mumbai portfolio hub.



From live prototype to institutional-scale risk infrastructure

Current Scope

Operates on pre-configured wholesale asset portfolios and throttled public REST API ingestion tiers.

Roadmap & Future Work

01 Scale ingestion throughput using Apache Kafka message queues.

02 Integrate fine-tuned open-source LLMs (e.g., Llama-3) for complex corporate entity reasoning.

03 Expand dynamic multi-asset index rebalancing.